



SYSTEM OVERVIEW

AUTOVET · VERSION 1.0

SYSTEM DESCRIPTION

AutoVet is a hybrid veterinary clinic management system that unites a cloud web portal, an offline-capable clinic workstation, and an AI-assisted inventory forecasting module into a single platform. Pet owners book appointments and manage pet profiles online, while clinic staff handle patient records, walk-ins, billing, and stock from one dashboard.

Built on a Laravel REST API with React front-ends, AutoVet keeps the clinic running even when connectivity drops and resynchronizes securely once it returns — replacing paper logbooks and spreadsheets with role-based, auditable digital records.

OBJECTIVES

- Automate appointment booking, scheduling, and clinic transactions.
- Centralize pet profiles, medical histories, and vaccination records.
- Prevent stockouts through FIFO batch tracking and AI demand forecasting.
- Generate invoices, receipts, and management reports instantly.
- Secure patient and owner data with role-based access and audit logging.
- Sustain clinic operations offline and sync reliably to the cloud.



SYSTEM FEATURES

WHAT AUTOVET DOES



Online Appointment Booking & Scheduling

Slot-based booking with double-booking prevention and vet schedules.



Pet Profiles & Electronic Medical Records

Species, breed, weight ranges, treatment history and vaccinations.



Inventory Management with FIFO Batch Tracking

Auto-deduction per transaction, expiry-aware batches, low-stock alerts.



AI Stock Forecasting

Predicts consumption and suggests reorder timing before shortages occur.



Automated Invoicing & Printable Receipts

Service and product billing with PDF invoice generation.



Reports & Analytics Dashboard

Sales, patient, and low-stock reports exportable to PDF.



Role-Based Access, Audit Logs & Notifications

Admin / Vet / Staff / Pet Owner roles with real-time alerts and backups.



TECHNOLOGIES USED

AUTOVET TECH STACK



React 19 • Vite • Tailwind CSS

Admin dashboard and pet-owner portal front-ends.



Laravel 12 • PHP 8.2 REST API

Application backend, business rules, and sync bridge.



MySQL / TiDB Cloud Database

Cloud database with a local MySQL mirror for offline use.



Laravel Sanctum • RBAC • Audit Trail

Token authentication, hashed passwords, per-role permissions.



Laravel Reverb • Pusher • Horizon

Real-time notifications and queued background jobs.



Python • Pandas • Scikit-Learn

Regression-based inventory demand forecasting module.



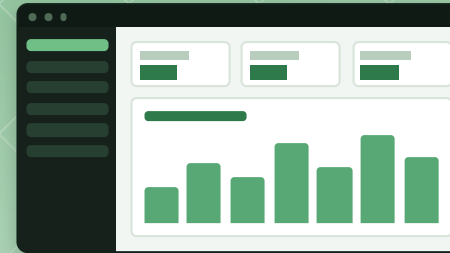
Render • Vercel • Supabase S3 Storage

Cloud hosting, static delivery, and media/document storage.

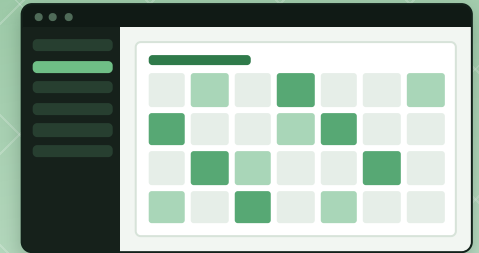


SYSTEM FEATURES

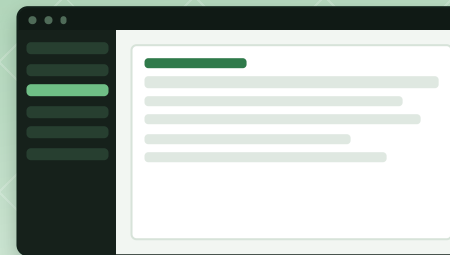
KEY SCREENS



1. ADMIN DASHBOARD



2. APPOINTMENT SCHEDULING



3. INVENTORY & FIFO BATCHES



4. INVOICING & REPORTS